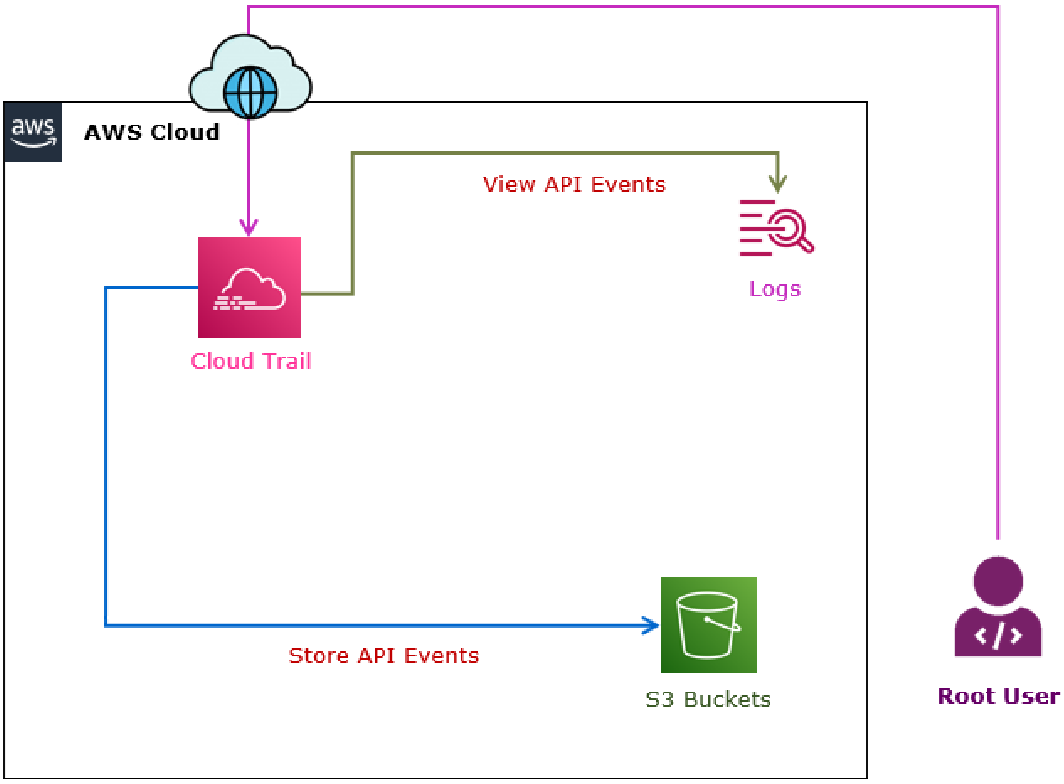
Optional Task: If time permits

Task 1: Store API Calls in S3 Bucket

In this task, you will store the trails into S3 bucket for compliance purpose and integrate the trail with CloudWatch to generate the alerts.



Step 1: Create AWS S3 buckets

1. In the **AWS Management Console**, on the **Services** menu, click **S3**.
2. Click **Create bucket**.
   a. **Bucket name**: Write **cloudtrail-123**.

Note: **Replace 123** with a random number to make bucket name unique.

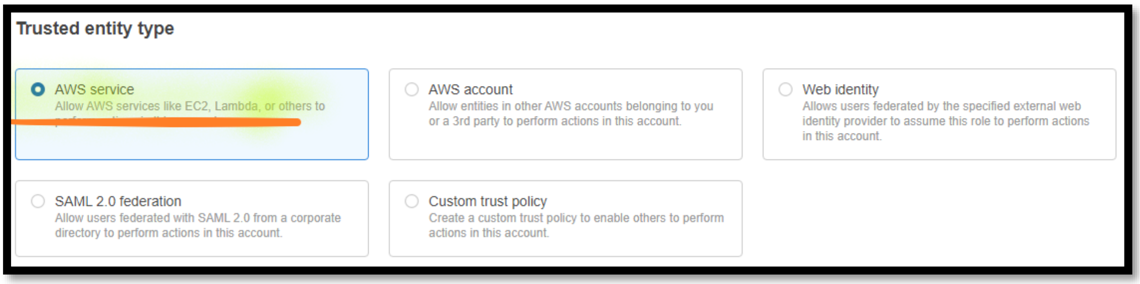
   b. **Region**: Dropdown and select **US East (N. Virginia)**.

Note: Leave other details as default.

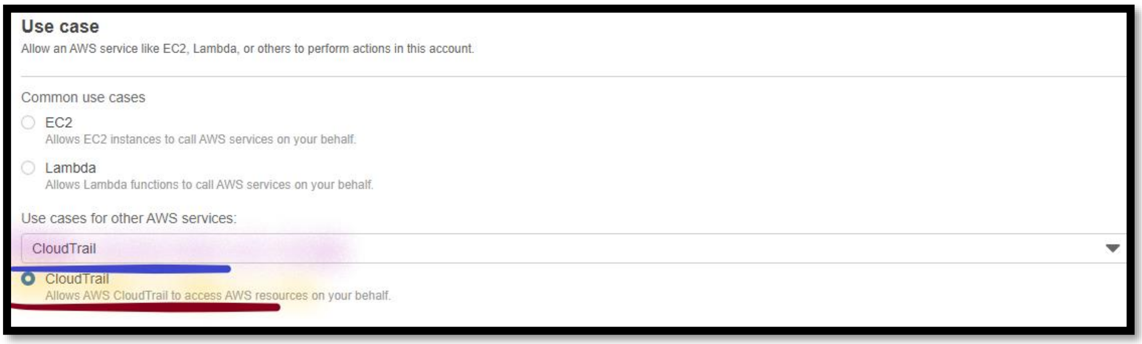
   c. Click **Create bucket**.

Step 2: Create IAM Roles for CloudTrail Logs

3. In the **AWS Management Console**, on the **Services** menu, click **IAM**.
4. Select **Roles**, click on **Create role**.
   a. In the **Select trusted entity** section:
      i. **Trusted entity type**: Select **AWS service**.



      ii. **Common use cases**: Dropdown and select **CloudTrail**.
         a) Select **CloudTrail**.
      iii. Select **Next**.



   b. In the **Add Permissions** section:

> **Note**: Leave all the details as default.

      i. Select **Next**.
   c. In the **Name, review, and create** section:

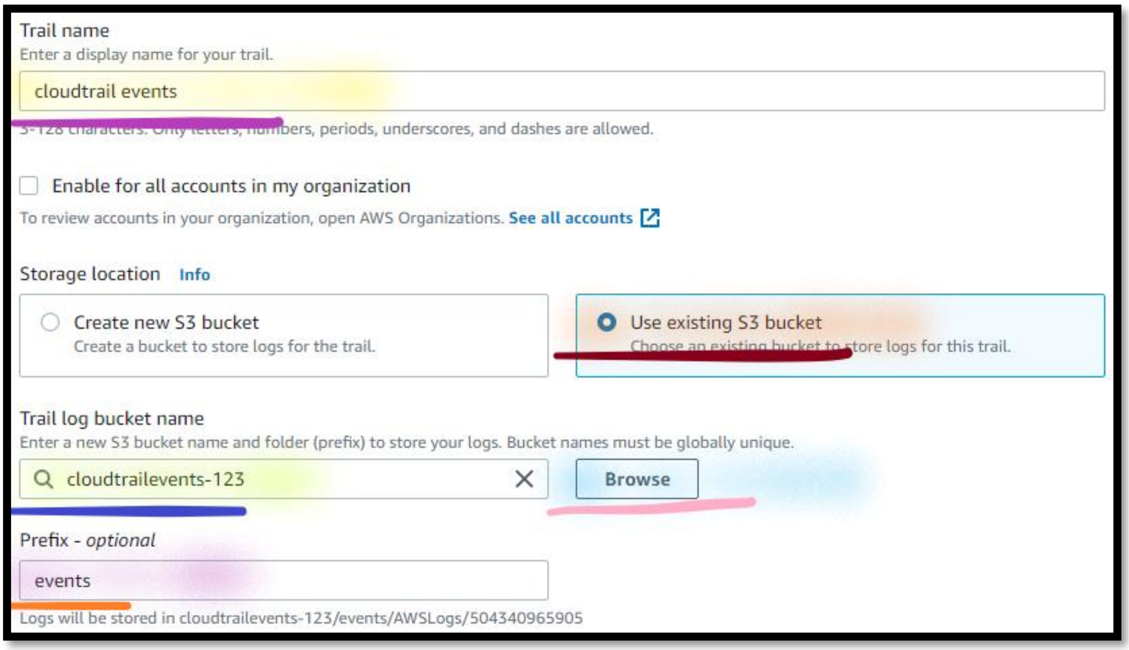
> **Note**: Leave all the details as default.

      i. Click **Create role**.

Note: **Wait**, till Role **AWSServiceRoleForCloudTrail** gets created.

Step 3: Enabling CloudTrail event logging for S3 buckets

5. In the **AWS Management Console**, on the **Services** menu, click **CloudTrail**.
6. Dropdown and select **US East (N. Virginia)** region.
7. Click **Trails**.
    a. Select **Create trail**.
        i. In the **General details** section:
            a) **Trail name**: Write **cloudtrail-events**.
            b) **Storage location**: Select **Use existing S3 bucket**.
            c) **Trail log bucket name**: Select **Browse**.
                1) Select **cloudtrailevents-123** bucket.
                2) Select **Choose**.
            d) **Prefix - optional**: Write **events**.



e) **Log file SSE-KMS encryption**: Uncheck the Checkmark against Enabled.

f) **Log file validation**: Uncheck the Checkmark against Enabled.

ii. In the CloudWatch Logs - optional section:

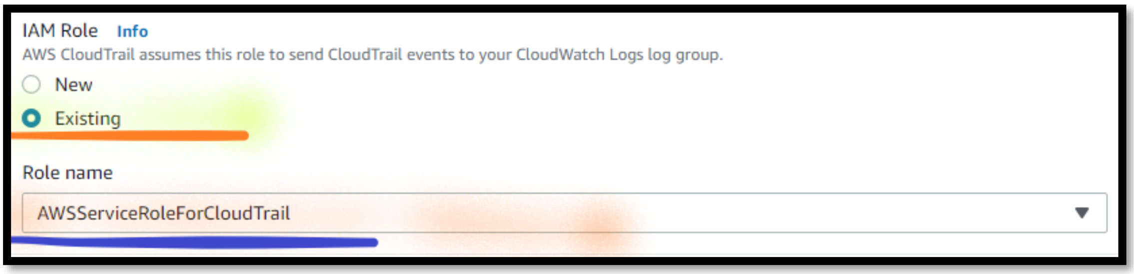
a) **CloudWatch Logs**: Enable the Checkmark against Enabled.

1) **Log group**: Select New.

2) **Log group name**: Write cloudtrail-logs.

b) **IAM Role**: Select Existing.

1) **Role name**: Dropdown and select AWSServiceRoleForCloudTrail.



c) Select Next.

iii. In the **Events** section:

> **Note**: Leave all the details as default.

a) Select **Next**.

iv. In the **Management Events** section:

> **Note**: Leave all the details as default.

a) Select **Next**.

v. In the **Review and create** section:

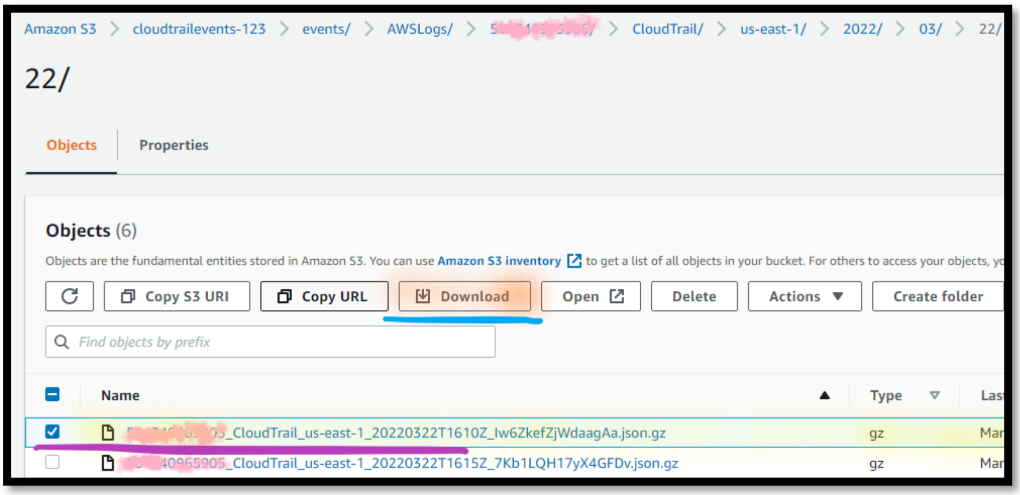
a) Select **Create trail**.

Step 4: View the events from S3 buckets

8. In the **AWS Management Console**, on the **Services** menu, click **S3**.
9. Select **Bucket**.
   a. Open **cloudtrailevents-123**.
   b. Open **events**.
   c. Open the **folders** and **subfolders** until you can see the **Object**.

> **Note**: You can see the Object name as "**account-name_cloudtrail_region_xxx**".

   d. Select the **object** and **Download**.



e. Open the **deliverystream-s3-xxxx** file in the **Notepad**.

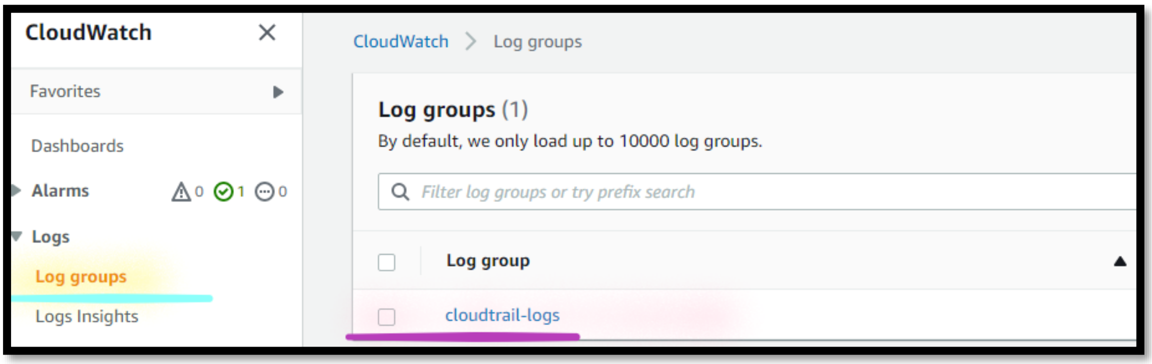
Note: You can see the **API details**.

Step 5: Monitor the CloudWatch Events

10. In the **AWS Management Console**, on the **Services** menu, click **CloudWatch**.

11. **Expand** the **Logs**.

a. Click on the **Log groups**.

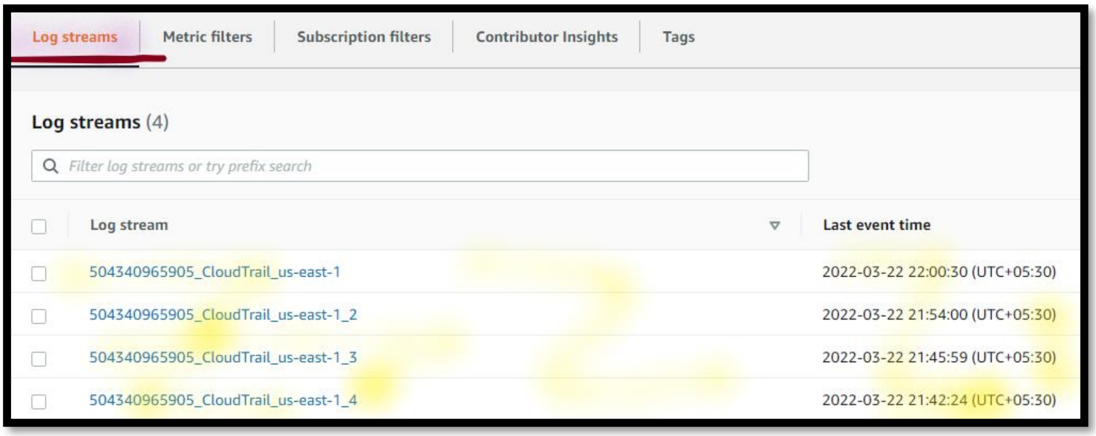


b. Open the **cloudtrail-logs** log group.

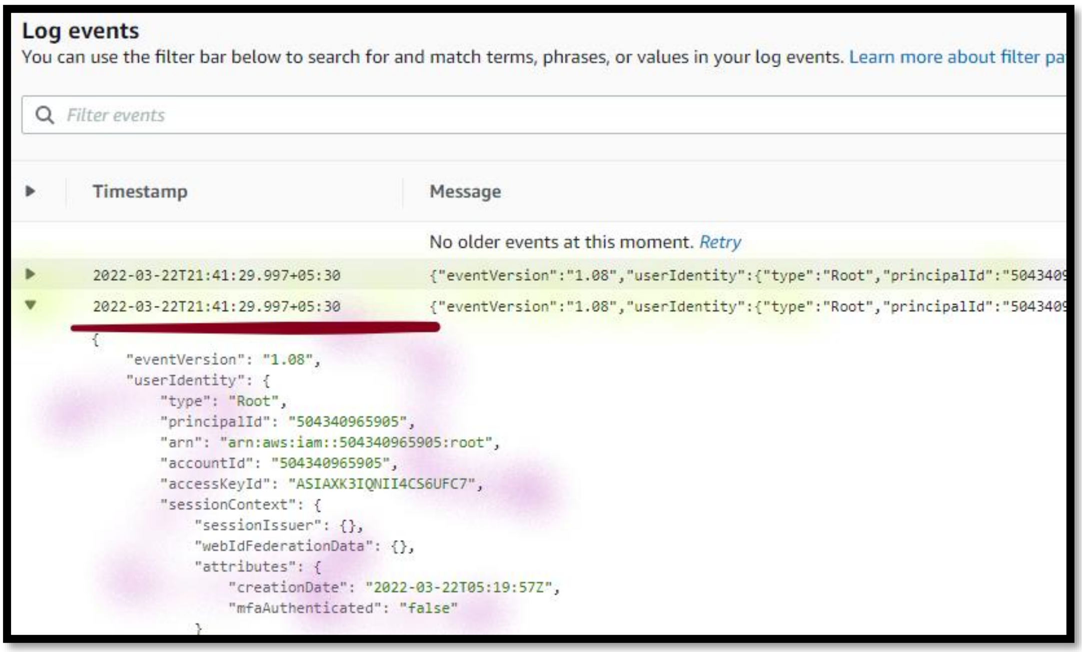
Note: You can see the **Log streams** generated by **Cloud Trail** with the **Date** & **Time**.

c. Select the **Log streams**.

i. Open the **latest** **Log streams**.



Note: **Inspect the contents** of all the recent **entries** of all the log streams.



Task 2: Cleanup the Environment

Step 1: Delete Trail

12. In the **AWS Management Console**, on the **Services** menu, click **CloudTrail**.

13. Click **Trails**.

    a. Select **cloudtrail-events**.

       i. Select **Delete**.

       ii. Select **Delete**.

Step 2: Delete Log group

14. In the **AWS Management Console**, on the **Services** menu, click **CloudWatch**.

15. **Expand** the **Logs**.

    a. Click on the **Log groups**.

    b. Select the **cloudtrail-logs**.

       i. Select **Actions**.

ii. Select **Delete log groups(s)**.

iii. Select **Delete**.

Step 3: Delete the S3 Bucket Object

16. In the **AWS Management Console**, on the **Services** menu, click **EC2**.

a. Select **cloudtrailevents-123** bucket.

i. Select **Empty**.

a) **Type** **permanently delete** to ***Delete all the objects*.**

b) Select **Empty**.

c) Select **Exit**.

b. Select **cloudtrailevents-123** bucket.

i. Select **Delete**.

d) **Type** **cloudtrailevents-123** to ***Delete the bucket*.**

e) Select **Delete**.